

Machine Learning - Assignment 1: Vehicle Price Prediction

Publication date: 23-April-2026

Due date: 14-May-2026

Data

In this exercise, you will work with the [Vehicle Price Prediction Dataset](#). The dataset includes various features related to automobile specifications, usage, and listing details. The goal is to predict either the **Condition** of the vehicle (Classification) or its **Price** (Regression).

Variables:

- **Brand / Model:** The manufacturer and specific model of the vehicle.
- **Year:** The year of manufacture.
- **Fuel_Type:** Type of fuel (Petrol, Diesel, CNG, Electric, etc.).
- **Transmission:** Manual or Automatic.
- **Mileage:** The total distance the vehicle has traveled (km).
- **Engine / Power:** Technical specifications of the motor.
- **Condition:** The physical state (Target for Classification).
- **Price:** The listing price in USD (Target for Regression).

Data Split

To ensure consistency, use the following row indices for your datasets:

- **Training set:** Rows 0 to 70,000 of the dataset (inclusive).
- **Validation set:** Rows 70,001 to 85,000 of the dataset (inclusive).
- **Test set:** Rows 85,001 to the end of the dataset (inclusive).

Section A - Coding (40 pts):

Goal: Implement a Decision Tree and Random Forest from scratch in Python.

- **Constraint 1:** If two features provide the exact same Gini Gain (for classification) or SSR reduction (for regression), your algorithm **must** prioritize the feature that appears first alphabetically.
- **Constraint 2:** Implement a `min_samples_leaf` parameter. Your code must discard any split that results in a leaf node with fewer than 5 samples.
- **Random Forest:** For the regressor, implement the average of the trees' predictions at test time.

Section B - Data Preparation (10 pts):

Goal: Explore and clean the data.

- **Visualization:** Explore the data using visualizations. The goal of this section is to get insights on the data which may or may not be relevant for the following sections.
Plots should have an informative main title, axis labels and a legend. For each, explain why you selected that specific plot type and what it reveals about the feature distributions. You are required to have a minimum of three plots.
- **Data Prep:** Some features have missing values, handle them accordingly and explain your choice of dealing with them.
- **Outlier Analysis:** Manually inspect the Price and Mileage columns. Use a statistical method (e.g., IQR or Z-score) to identify outliers in Price and Mileage.
- **Artifact Analysis:** Investigate the "Floor Effect" in the Price column (where 5% of values are exactly \$1,500) and the "Clumps" in Mileage (e.g., exactly 300,000 km). Explain if these are true outliers or artifacts of data acquisition, and justify your decision to keep or remove them.
- **Justification:** Provide a short paragraph explaining your specific threshold for an outlier and why you chose it. Did removing these entries change your data distribution significantly?

Section C - Implementation & Complexity (30 pts):

Goal: Train your scratch-built models from **Section A (4 models)** and analyze their performance.

- **Training:** Perform both classification (to predict Condition) and regression (to predict Price).
- **Complexity Curve:** For your Decision Tree Regressor, generate a plot showing **Training MSE vs. Validation MSE** for `max_depth` values ranging from 1 to 20.
- **The "Elbow" Analysis:** Identify the exact depth where your model begins to overfit. Based on this plot, explain which `max_depth` you would choose for a production model and why.

Section D - Comparison (10 pts):

Sklearn Models

Implement the 4 models (DT and RF for classification and comparison), including hyperparameter tuning using the built-in functions from the sklearn library.

Comparison

1. Compare the results of your program and the built-in sklearn models in terms of metrics and runtime. If there are large differences, suggest an explanation to why.
2. Compare the results of the random forest regressor (the sklearn model you're your implementation) when using average and median at test time, comment on the differences.

Section E - Interpretation (10 pts)

- **Feature Importance:** Extract and visualize the feature importance from your **Random Forest**.
- **Model Reasoning:** Which feature was the strongest predictor of price? Does this align with real-world intuition (e.g., does Year matter more than Mileage)?
- **Failure Analysis:** Identify one specific vehicle in the test set where your model's prediction was significantly incorrect. Suggest a logical reason why the model might have failed for this specific case
- Use your **Decision Tree** to compute the feature importance, did you get the same important features, explain your results

Section F - Bonus (20 pts)

- Classification Metric
Report the sensitivity and specificity metric of section C. Is there a significant difference between the scores? Suggest an explanation as to why this may be the case. Suggest a method to improve the score.
- Random Forest Regression Test
Change your implementation of testing the random forest regression to median (section A), rerun the implementation in Section C. Compare the new results with using the average. Discuss the changes and reflect on them.
- Algorithmic Optimization
Implement a version of your Decision Tree that uses a **Histogram-based split** or **Feature Pre-sorting**. Instead of evaluating every possible threshold for continuous variables at every node, bin the data into discrete intervals to speed up the Gini/SSR calculations.
 - Compare the training runtime of this optimized version against your Section A implementation.
 - Discuss whether the loss of precision (due to binning) significantly impacted the accuracy.

Submission Guidelines

Please read the following section carefully before submitting the assignment.

Coding Guidelines

- Each model should be trained on the training and validation data.
- Each model should predict the label of the test data and report the following measures, you can use the built-in functions in Sklearn.
 - Accuracy for classification.
 - MSE for regression.
- Impurity measures are Gini for classification and SSR for regression.

- Use familiar packages with explicit explanations. Except for parts where you were explicitly asked to use the Sklearn library, do not use this library or any other machine learning library).
- Try to minimize the usage of loops, lists and other inefficient programming habits. Numpy library has a lot of useful built-in functions. In this case, consult with the library documentation and your chosen search engine.
- If you have installed any libraries beyond those presented in the exercises, please specify this in the report.
- The code should run without warnings or errors.
- Good documentation is critical.
- Indicate the exercise sections in the code as well.
- Use meaningful variable names.
- Do not use reserved words.
- Use constants where possible.

Submission Guidelines

- The assignment should be submitted in pairs (only one submission).
- You are required to submit two files including all the sections. One in **.ipynb** format and one in **.html** format. **Both files should also include the program's outputs.**
- The files' names should be of the form: ML_HW1_ID1_ID2.
- Assignments submitted late will receive a penalty of 3 points for each day, up to one week. Later submission will not be accepted.

Grading

You can get more than 100 points for the exercise. The exercise will be graded according to the following criteria:

- Correctness.
- Clarity.
- Efficiency of implementation.
- Elegance of implementation.

Self-learning

- As we mentioned at the beginning of the course, self-learning is an important part of the course. Treat all sources of information carefully and critically.
- Usage of LLMs is allowed, but reference it when it was used.
- You can and should consult with other students in the course, but each pair must write their own work.
- It is reasonable to assume that not all results and algorithms will be identical.

Questions and Reception hours

- Please post your questions on the exercise forum in Moodle, after you have read the previous posts. Professional questions sent by email will not be answered.
- If you want to schedule a reception hour with one of the instructors - please send your questions by email in advance.
- In any other case (personal questions, request for an extension with a justified reason, etc.) please email the instructor.

Good luck